

# How promoted tasks and services rank

A Meta-style ad auction: bid × action rate, second-price billing, paced daily budgets. 21 September 2026.

A poster or worker can pay to push a listing to the top of the feed. The model is the one Meta uses: the highest bidder does not simply win, the budget is genuinely spent over the day, and delivery stops when it runs out.

## 1. STARTING A CAMPAIGN

The Promote screen takes a daily budget and a duration, which multiply into the total charged up front.

```
amount_minor = daily_budget × days    (paise)
```

- `start_promotion()` creates it as pending — checks the listing is yours and refuses a second live campaign on it, so a listing cannot bid against itself.
- `activate_promotion()` needs a payment that has actually settled. That is the gate between "opened a payment page" and "is being advertised".
- Activation stamps the window: `starts_at = now`, `ends_at = now + days`.

## 2. ELIGIBILITY: A WINDOW, NOT A FLAG

```
status = 'active' AND starts_at ≤ now() < ends_at AND spent_today < daily_budget
```

Sponsorship is derived, never stored. Placement therefore ends on time by itself — no nightly job to run, and none that can fail and leave a finished campaign still sitting at the top of the feed.

## 3. THE AUCTION: BID × ESTIMATED ACTION RATE

This is the part that makes it Meta-like rather than a rich-list. Ranking multiplies the bid by how likely people are to act on the ad, so an ad nobody taps cannot buy its way to the top for long.

```
total_value = daily_budget × eCTR × pacing_factor
```

eCTR is clicks ÷ impressions, smoothed toward a 5% prior over the first ~50 impressions. Without the smoothing, a campaign with one impression and one lucky click would read as a 100% click rate and win everything.

**Measured result — the low bidder wins:**

- RANK 1 — bid ₹80/day, eCTR 0.250, total value 2000, pays 2160.
- RANK 2 — bid ₹300/day, eCTR 0.018, total value 540, pays 1 (the floor).

The ₹80/day campaign outranks the ₹300/day one because people actually tap it. Under the old logic the ₹300 bid won automatically.

## 4. SECOND-PRICE BILLING

The winner pays what it took to clear the runner-up, not their own bid — capped at their bid, floored at one paisa.

```
charge = min( own_bid , next_total_value ÷ own_eCTR )
```

In the table above the winner bid ₹80/day but was charged 2160 (540 ÷ 0.25). Second price is why bidding your true value is safe: bidding higher cannot make you pay more than the competition demands.

## 5. PACING AND THE DAILY BUDGET

A daily budget is now real money that depletes. Each impression is billed a slice of the auction CPM:

```
impression cost = charge ÷ 1000    (minimum 1 paisa)
```

- Spend is summed per campaign per calendar day.
- A campaign ahead of schedule is damped, so a day's budget is not gone by mid-morning.
- When `spent_today` reaches the daily budget the campaign leaves the auction and returns tomorrow.

Verified end to end: a ₹80/day campaign served exactly 8,000 impressions at 1 paisa, spent exactly 8000 paise, never overspent, and dropped out of the auction at exhaustion.

## 6. DELIVERY AND REPORTING

Every serve and tap is recorded in `ad_events`, which is what both the auction and the advertiser's reporting read.

- Impression — billed, logged when a sponsored card reaches the screen, once per task per feed load.
- Click — free, but it feeds the eCTR that decides future ranking.
- `my_campaign_stats()` returns impressions, clicks, CTR and spend for your own campaigns.

## 7. WHERE IT LIVES

- `task_promotions` — one row per campaign (budget, days, window, audience).
- `ad_events` — impressions and clicks, with what each one cost.
- `ad_auction()` — the live auction: ranks, prices, drops exhausted campaigns.
- `record_ad_impression()` / `record_ad_click()` — delivery, called by the feed.
- `my_campaign_stats()` — advertiser reporting.
- `settle_finished_campaigns()` — hourly sweep: refunds and books revenue.

## 8. SETTLEMENT: YOU ONLY PAY FOR DELIVERY

The campaign total is collected up front, but only the delivered part is ever earned. When a campaign's window closes — or the advertiser stops it early — it is settled in one transaction:

- The unspent remainder is returned to the advertiser's in-app wallet, with a `wallet_adjustments` row as the audit trail.
- The delivered spend is booked to `platform_ledger` as `ad_revenue`, alongside task commission and poster fees.
- `settled_at` is stamped, so a campaign can never be settled — or refunded — twice.

Cancelling mid-flight refunds immediately rather than waiting for the sweep; that is precisely the case where money is owed back. A cron job sweeps finished campaigns hourly, because a refund that waits until the small hours is a refund the advertiser notices is late.

**Measured: money is conserved**

- ₹200 collected, 300 paise delivered → ₹197 refunded + 300 paise revenue = ₹200 exactly.
- Running the sweep a second time settled nothing: no double refund.

**OPEN DECISION**

The refund goes to the in-app wallet, not back to the original card or UPI. That is deliberate — it matches how disputed escrow is returned, and Razorpay cannot presently refund because the account is on a test key and not activated. If advertisers should be repaid to source instead, that is a change to make once payments are live.

`activate_promotion()` still requires a settled payment, and Razorpay cannot collect one yet: the account is on a test key and not activated. Campaigns can be created but not activated by a real user until that is resolved. The auction itself was verified by activating campaigns directly in the database.